* ARM Topics *

1. Fundaments (ARM.V7 peocessoe, pipeline, modes, eegistee).
2. Delay Geneeation using timees.
3. GPIO peogeamming (LED, Push button, Alpha LCD)
4. UART peotocol (Bluetooth, RFID)
5. ADC (sensoes, inteefacing, ADC to Human, Analog inteefa
vimp 6. Inteeeupt handling.

7. SPI peotocol
8. I2C peotocol
9. CAN peotocol (← Robeet Bosch is invented)

CAPL sceipting. } online Teaching
RTOS (Real time O.S)
↳ It is used in haedwaee.

* Fiemwaee engineee :-
  - It is also called as 'embedded engineee'.
  - The peogeam which is without O.S. it is called 'Fiemwaee'.
  - embedded-C.
  - μC. peogeam.

keil ← It is inbuit compilee
IDE ↑ Integeated Development Envieonment

* BSP – Boaed Suppoet Package

* Test Patteen.

ARM WT1 (30 m)
ARM. WL1 (20 m) ✓
ARM. MT (50 m)
ARM. ML (50 m). ✓

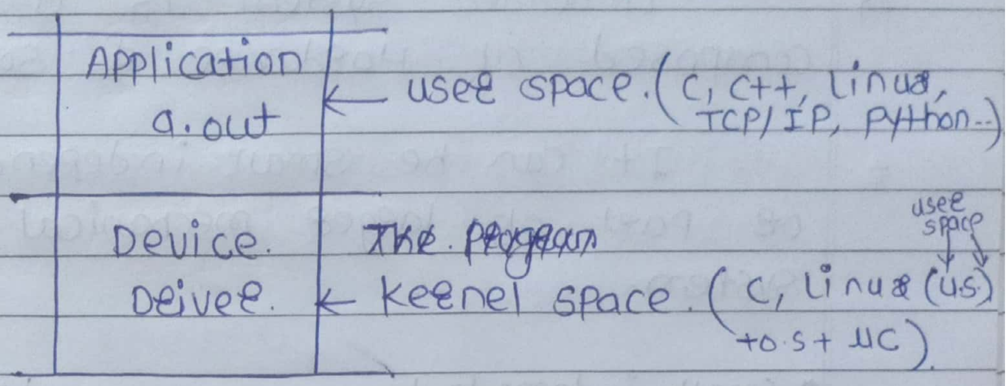
Aeduino → It is a Conteollee.

01/04/2024

* ARM *

a.out $\xrightarrow{\text{Before loaded}}$ Hard disk.

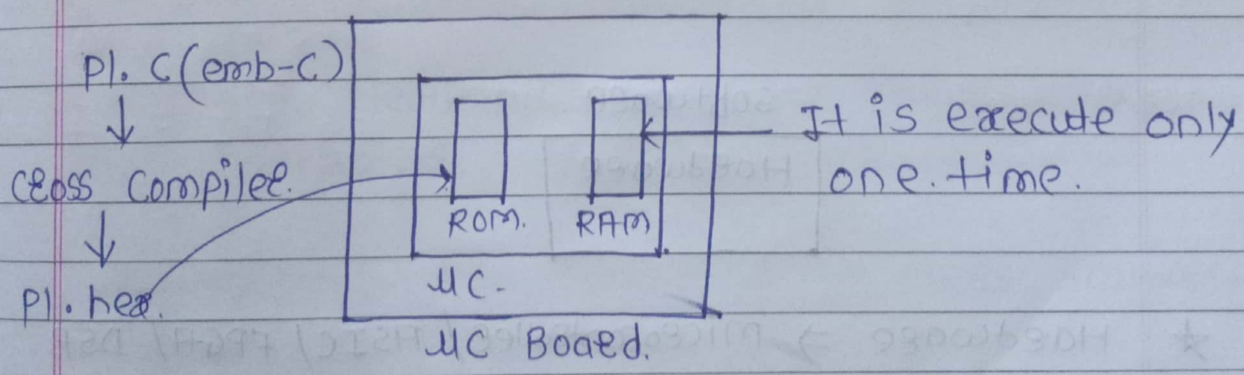
./a.out $\xrightarrow{\text{After loaded}}$ RAM.



- The program executes in user-space. it is called as "Application".
- The program executes in kernel space. it is called as "Device-Drivers".

* Microcontroller chip :- NXP semiconductor.

- It is stored only one program.



- P1.hex File stored in ROM.
- Transfer the hex File into ROM, it is called "Flashing / Dumping hex File".
- chip manufacturers replaced ROM with Flash memory.

02/04/2024.

*. Firmware / Bare-metal code / Raw code / Emb-C code

Q. What is embedded-system?

⇒ Embedded system is a system composed of Hardware & Software.

- It can be small independent system or part of larger mechanical & electrical system.

* Small independent	larger
Remote	Dish-washer
Bluetooth headset	washing machine
keyboard & mouse	Refrigerator
Smart watch	AC, automatic door in metro
	GPS system

☆ Analog watch not comes in embedded system.

Software

Hardware

☆ Hardware ⇒ microcontroller / ASIC / FPGA / DSP.

- Microcontroller :- It is used for general purpose application.

- ASIC :- Application specific Integrated circuit
  - It is designed for specific application.
  - It is faster than microcontroller.

- FPGA :- Field programmable Gate Array.

  - It is customize to execute in another application.

- DSP :- Digital Signal processor

Software.

↓

Program.

embedded system without O.S	embedded system with O.S.
embedded-C.	C-language.
embedded-CPP.	CPP.
Assembly.	Linux.
(Basic programming Language.	Python scripting.

* Major application areas of emb-system :-

- Consumer electronics :- smart phones, TVS, Gaming Tablets, smart watches ...

- Automation :- Fax machines, printers, scanners.....

- Telecom :- Routers, Switches, cellular phones...

- Military aerospace :- satellite system, sonar, radar.

- Automotive electronics :- ECU'S, infotainment system...

- Industeial conteol : smaet sensoes ...

- Remote automation : Heating - ventilation, Aie-conditioning (HVAC), utility meters ...

- medical electeonics : ECG machine, city scan machine, oximetee, Glucose metee, Theemometee ...

☆ Classification of an Embedded-System :-

System I	System II	System III
MCU based, Veey low powee, Small code (KB's), Baeem etal, small RTOS (upto 100 MHz)	mpu based. High speed. (~~200 MHz~~ till 1GHz). O.S. + Application code.	MPU based special copeocessoes, veey High computat powee, special Haedwaee. acceleeate engines like TPU, VPU, GPU.
e.g. weist Band	e.g. Smaet-watch.	e.g. AI Smaet watch.

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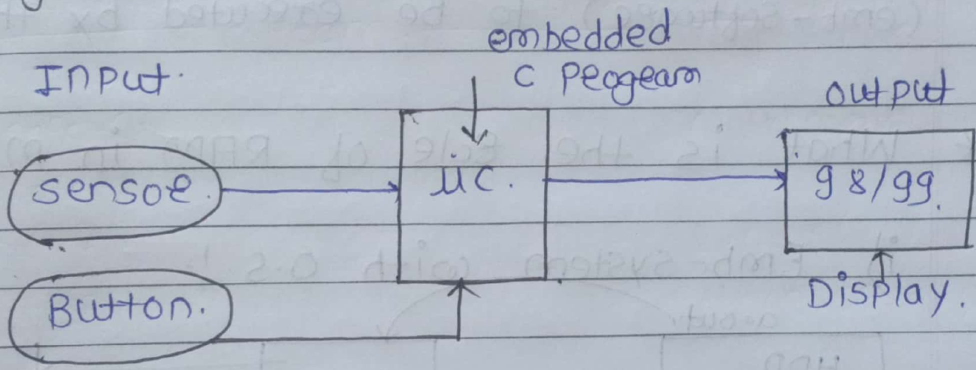
* Board Bringup → customization of linux, Installation of new linux O.S.

* What is emb-system consists of?

⇒ - I/P device
- Microcontroller (The Brain)
- O/P device.

e.g. Pulse oximeter.



* Microcontroller :-

- µC is a programmable chip which consist of internal memory, peripherals and processor.

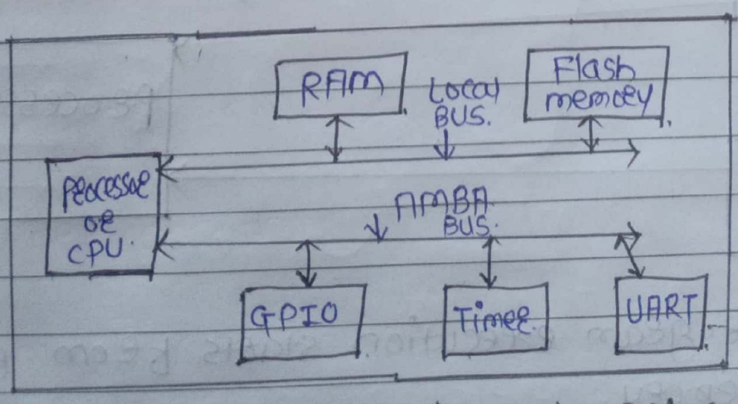


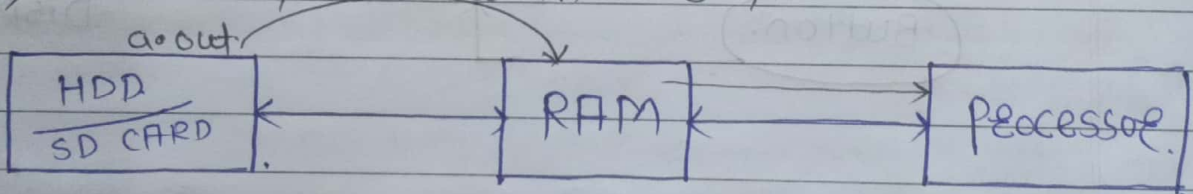
Fig. Internals of Microcontroller (ARM)
OR
Block Diagram of µC.

- 8051.
- AVR - Advanced Virtual Risk.
- PIC - Peripheral Interface Controller.
- ARM - Advanced RISC machine.
- AMBA Bus - Advanced microcontroller Bus architecture [It connect peripheral to processor].

* Flash memory :-
  - It is also called as code memory / program memory.
  - Flash memory will store emb-c program (emb-software) to be executed by the processor.

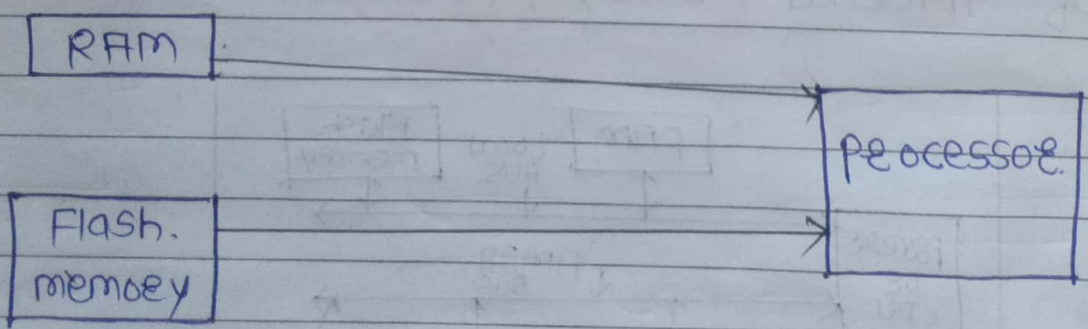
* What is the role of RAM in microcontroller?

i) Emb-system with O.S. :-



- Program execution starts from RAM.

ii) Emb-system without O.S. :-



- Program execution starts from Flash memory.

Imp.
* RAM is used to store runtime data.

* **Processor / CPU :-**
- Processor will execute a program which is stored in the Flash memory.
- It has not have on-chip memory.

* **Peripherals :-**
- one peripheral cannot communicate to each other.
- Through processor it will communicate.

• microcontroller is also called small computer in single chip.

Microcontroller (μC)	Microprocessor (μP)
1) μC is a programmable chip which comes with on-chip processor, memory & Peripherals.	1) μP is a chip which comes on-chip program execution logic units.
2) μC contains on-chip memory and peripherals.	2) μP doesn't contains on-chip memory & peripherals
3) μC are designed for embedded system	3) μP are designed for general purpose computers.
4) power consumption is less.	4) power consumption is more.
5) e.g. 8051, AVR, PIC & ARM Controller	5) e.g. 8085, 8086 it is designed by intel; i3, i5, i7 ... e.g. it is used in computers & laptop

04/04/2024.

☆ Different type of CPUs :-

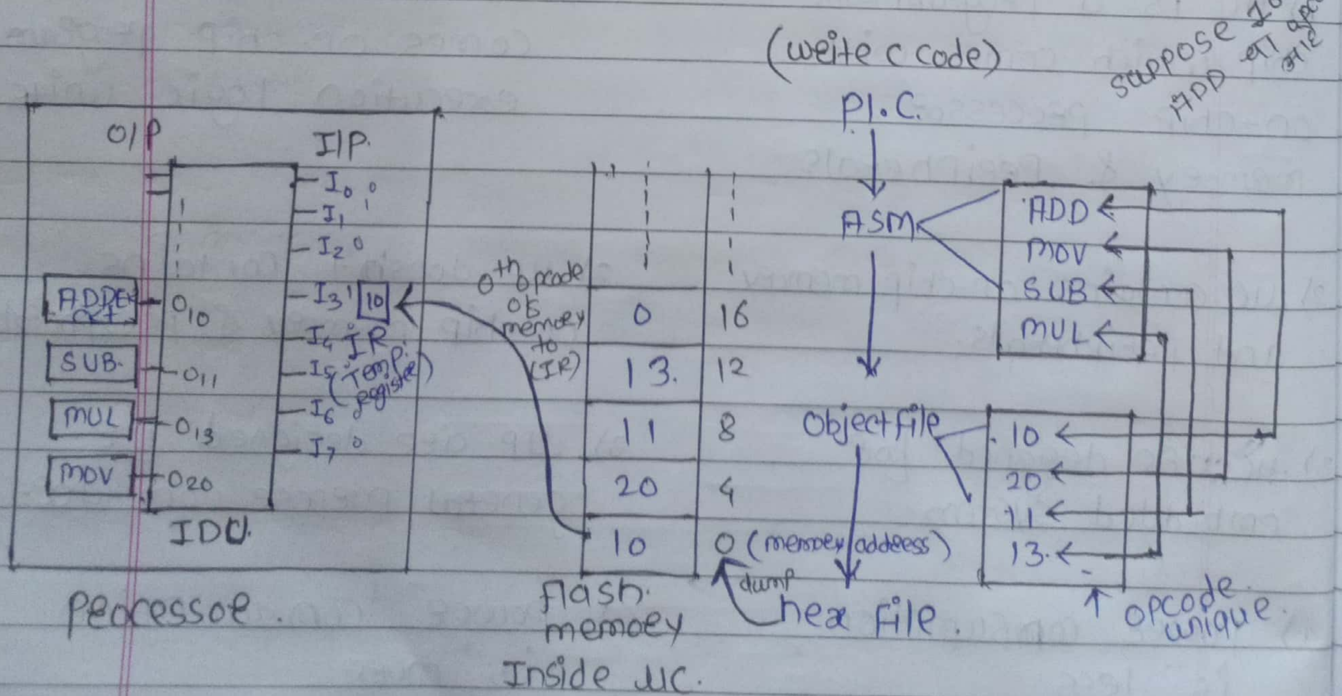
1. Based upon instruction set.

   CISC & RISC.

2. Based upon memory organization.

   Harvard & Von-neumann

- Every microcontroller has their own assembly instruction set & syntax of assembly is different.
- Assembly language is Hardware dependent language.
- Assembly language is not Portable.
- Bias Project written in Assembly language.



☆ IDU :- Instruction Decoding Unit

☆ IR :- Instruction Register

* Previous Diagram Explanation *

- ASM File is consists of an assembly instructions.
- object File is consists of an instruction opcode.
- opcodes are stored in Binary format.

Note: Every assembly instruction has its unique instruction num called as "opcode".

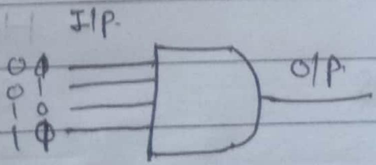
- The processor will fetch 0^{th} opcode memory to IR.
- processor will give IR to IDU.

suppose. 16 Bit (RISC)	250 Bit (CISC)
- 16 Instructions required.	- 250 instructions required
- 16 opcodes required.	- 250 opcodes required
- 16 logic gates are required	- 250 logic gates required.
- less power consumption	- High Power consumption
- less logic gates.	- more logic gates

(This is for only understand the CISC, RISC concept).

★ every logic gate takes voltage level as I/P.

suppose i) AND Gate.



A	B	O/P
0	0	0
0	1	0
1	0	0
1	1	1

$1 \rightarrow$ 5Volt

$0 \rightarrow$ 0 Volt

XCH, XCHD ← It reduce the code of lines but increases the complexity of logic gates

graphikaa Date Page

☆ Suppose e.g.

MOV d, S (data of source copied into destin

MOV R0, R1 (data of R1 is mov in R0)

Suppose we swap two numbers using MOV

e.g. R0 = 10, R5 = 20, R1 ← (temp. variable)

MOV R1 R0 (10) O/P: $R_0 = 20$, $R_5 = 10$

MOV R0 R5 (20)

MOV R5 R1 (10)

XCHD	R0	R5
	12	49

only last nibble swap.

Suppose we swap two numbers using XCH

XCH R0, R5 (all data will swap)

Using MOV	Using XCH.
- It increases code size (lines)	- It reduces code size (lines)
- It has less no. of logic gates	- It has more no. of logic gates.
- Hardware (H/W) complexity reduces	- Hardware (H/W) complexity reduces.
- simple instructions	- complex instructions
- Processing becomes less	- Processing becomes more.

CISC	RISC
1) complex instruction set computing	1) Reduced instruction set computing.
2) CISC will have more instructions and complex instructions.	2) RISC will have less instruction & simple instruction.
3) Agenda / purpose : To reduce no. of instruction per program.	3) Agenda / purpose: To reduce no. of clock pulses per instruction
4) No. of logic gates are more.	4) No. of logic gates are less.
5) power consumption is more and that's why heat discipation is more.	5) power consumption is less & that's why heat discipation is less.
6) more addressing modes in the ASM programming.	6) Less addressing modes in the ASM programming.
7) e.g. All intel processor...	7) e.g. ARM, AVR & PIC...

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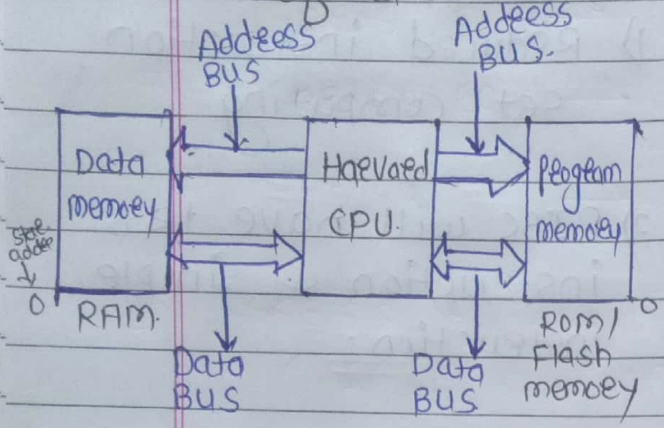
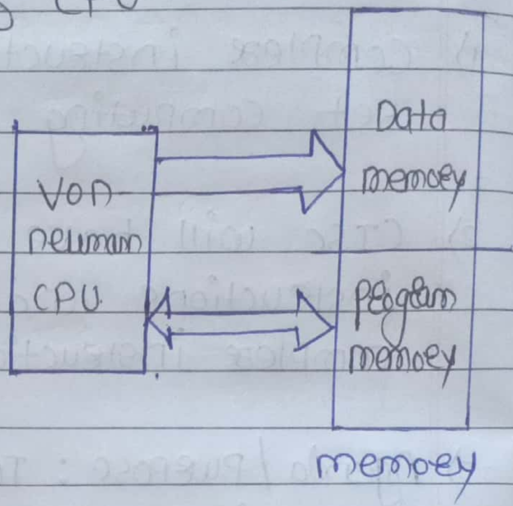
Harvard Architecture of CPU



Von-Neumann Architecture of CPU.



ii) Hardware complexity Increases.	ii) Hardware complexity reduces
iii) It process fast	iii) It process slow
iv) Address Bus decides the maximum limit.	iv) Here also address Bus decides the maximum limit.
v) e.g. 8051-CPU, AVR-CPU, PIC-CPU...	v) e.g. ARM7, 8085, 8086.

Imp

* If address Bus is 16 bits the memory limit is $\boxed{2^{16}}$

- Using address Bus & data Bus interface the memory.

- Addr Bus will always go towards memory.

☆ LPC2129 Microcontroller

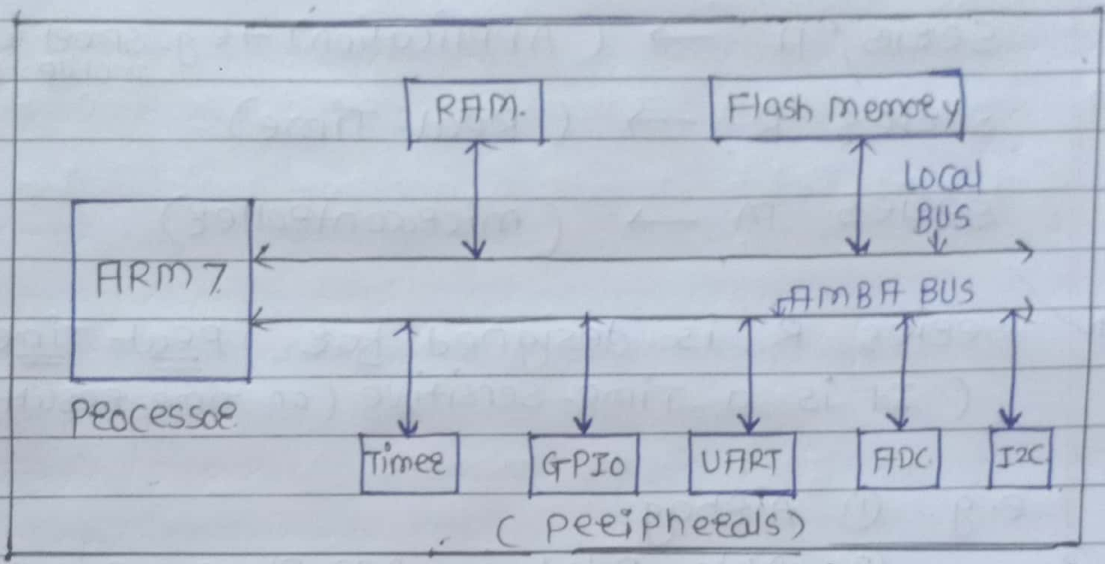


Fig. LPC2129 μC (NXP)

- ARM7 Processor is used in any controller. it is called " ARM 7 Controller "
- LPC2129 μC. is designed by NXP semiconductor
- Advanced RISC machine (ARM)
- Designed by arm Ltd company.
- Evolution of arm processor is arm.
  - arm1
  - arm2
  - arm3
  - arm6
  - arm7
  - arm8
  - arm9
  - arm10
  - arm11

☆ after arm11 company started.

☆ ARM Cortex :-

ARM Cortex Series :-

Series A → (Application) → e.g. Smart watches, mobile phones.

Series R → (Real-Time)

Series M → (Microcontroller)

☆ Series R is designed for "Real-Time-System"
(It is a Time-Sensitive. (on-time result))

e.g. ① Airbag
② Auto-pilot in Aeroplane or Autocar
③ Missile.

☆ Non-real Time :-
→ e.g. ATM, Lifts, washing machine.

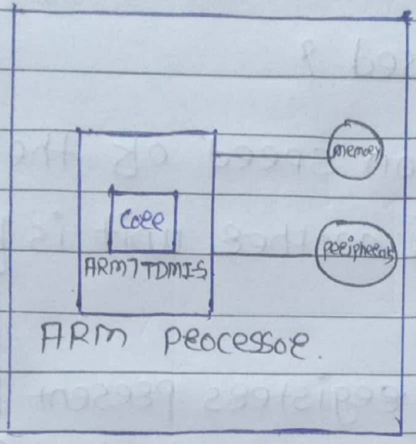
☆ Series M :-
e.g. :- Battery operating system as well as Non-real time. systems.
e.g. Smart watch.

☆ LPC2129 Microcontroller :-

Processor : ARM7 (unicore) — one core.

Core name : ARM7TDMI-S

Architecture : arm V4T



LPC2129 µC.

Q. What is the role of core?

⇒ core is used to execute a embedded program line-by-line (means one instruction at the same time)

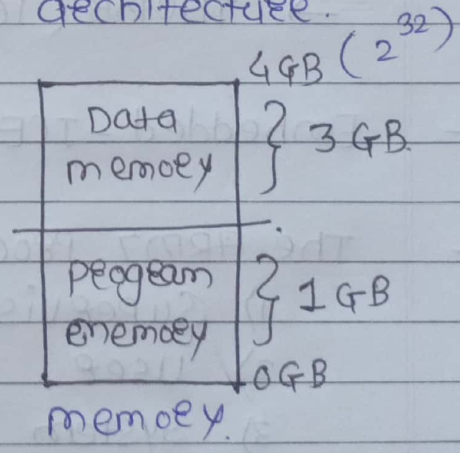
☆ Features of ARM7 Processor :-

- Low power & High performance processor.
- It works on 3.3 V & operating freq. 60 MHZ.
- It is 32-bit RISC processor.
- It supports Von-neumann architecture.
- The memory split ratio is $\boxed{3:1}$

3 : 1

3 GB for Data memory

1 GB for program memory.



- It supports 3 stage pipeline architecture.

  ⁕. 3 stages are.
  - → Fetch
  - → Decode
  - → Execute.

Q Why pipeline is used ?

⇒ To increase execution speed of the processor
(1 unit Fetch decode & another unit is for Fetch & execution)

- 37 Core registers (registers present in core)
  (a) registers are used to store data & address.
  b) It is temp. memory)

- processor is a small memory that is nothing but register.

- ARM7 supports diff type of instructions.
  → ARM (32 bit instruction set)
  → THUMB (16 bit instruction set)

- Embedded-ICE & on-chip debug logic.

- The ARM7 processor has 7 modes of operation
  1) Supervisor
  2) User.
  3) System
  4) FIQ
  5) IRQ
  6) undefined
  7) Abort

Scanned with CamScanner